

Variables & Data types

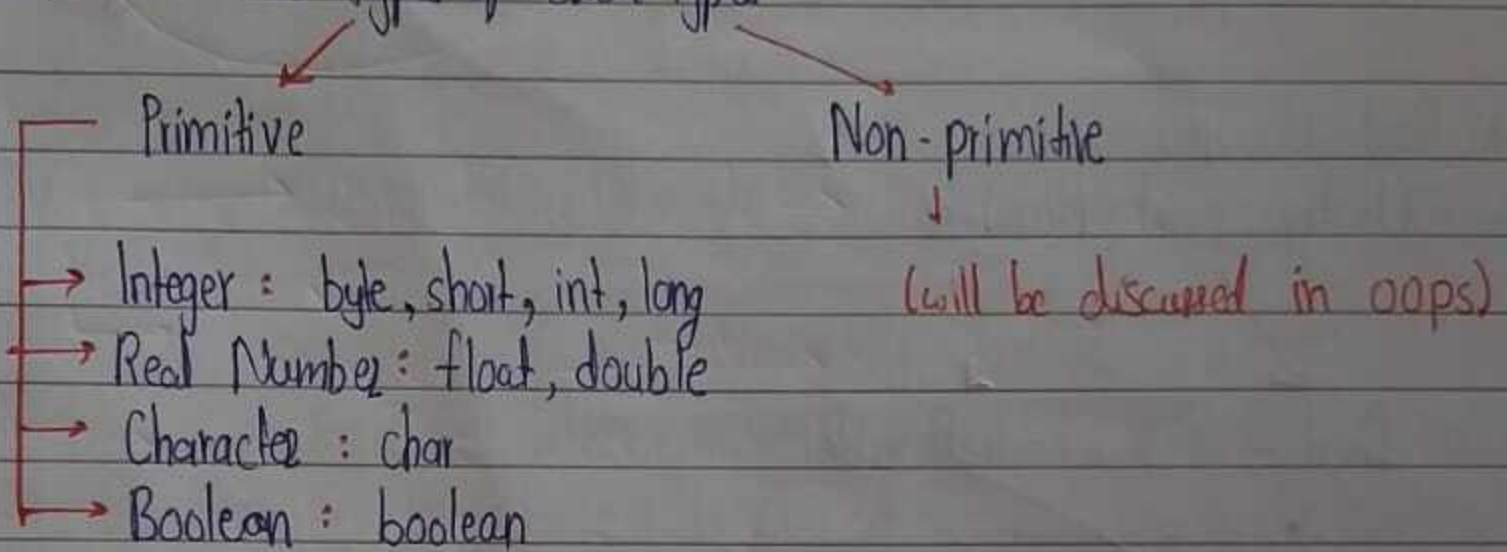
* Variables

→ Variables are containers which can store some value in it.

eg: `int firstNum = 10;`
 Annotations:
 - `int` → Data type
 - `firstNum` → identifier
 - `=` → assigned to
 - `10` → value
 - `10` is also labeled as *literal*
 # java is statically typed language

* Datatypes

→ There are two types of data-types



* Integer data types

→ used to store integers

eg: `_____ firstNum = 10;`
 Annotations:
 - `byte`
 - `short`
 - `int`
 - `long`

all have diffⁿ ranges

∴ it is used based on range of value we are storing

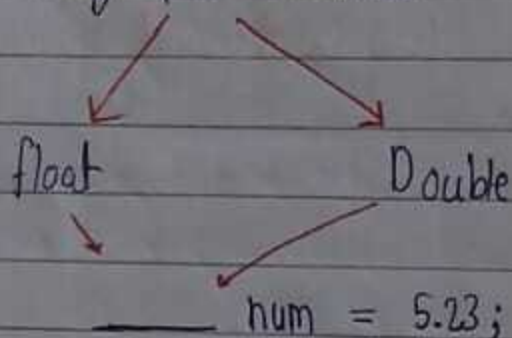
Name	Width (in bits)	Range
long	→ 64	→ -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
int	→ 32	→ -2,147,483,648 to 2,147,483,647
short	→ 16	→ -32,768 to 32,767
byte	→ 8	→ -128 to 127

Every number is signed in java
(+ or -)

∴ for a data type having width = b
Range = -2^{b-1} to $2^{b-1}-1$ } only for integer data types

* Real Numbers

→ floating point Numbers



Depends on range that which data type will come

Name	width (in bits)	Range
double	→ 64	→ 4.9×10^{-324} to 1.8×10^{308}
float	→ 32	→ 1.4×10^{-45} to 3.4×10^{38}

float → single precision (rarely used)
double → double precision (we prefer double in our projects)

* Characters

→ char : unicode character representation

ASCII = a-z

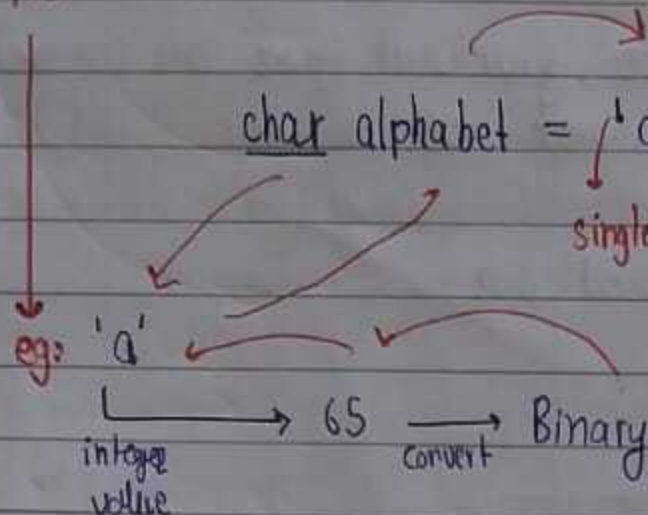
A-Z

0-9

- ; ^ " ! , ...

} before java, it was used

all characters
are represented
in ASCII format



char alphabet = 'a';

single quote represent it is character

but java used unicode instead of ASCII as ASCII used 8 bits
∴ char → 16 bits

* Boolean

→ boolean : stores true or false

(1) ^{in binary} (0)

boolean isTrue = False;

* Declaring & Defining

int x = 5; declaring & defining at the same time

int x; → declaration

x = 5; → Definition